Zhiyuan Chen

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Professor
Business School, **Renmin University of China**
59 Zhongguancun Street, Haidian District
Beijing 100872, China

Nationality: China
Email: chenzhiyuan@rmbs.ruc.edu.cn
Website: zhiyuanryanchen.github.io

Employment

Professor Renmin University of China, Business School 2026–present
Assistant Professor Renmin University of China, Business School , 2020–2026

Education

Ph.D. Economics, **Pennsylvania State University** 2015–2020
M.A. Economics, **Renmin University of China** 2013–2017
B.S. Economics (with distinction), **Renmin University of China** 2009–2013

Research

Research Fields

Industrial Organization; Development; International Trade
Particular interests: technological change, innovation, and industrial policy.

Publications in English Journals

[1] "Identifying Treatment Effects on Productivity: Theory with an Application to Firms' Production Digitalization," with Moyu Liao and Karl Schurter, ***RAND Journal of Economics***, forthcoming.

[2] "CO_2 Emission Regulation and Generation Allocation with Heterogeneous Coal-fired Generators," with Rong Luo and Li Su, ***Journal of Development Economics***, 2026, Vol. 180, 103771.

[3] "A Cost–Benefit Analysis of R&D and Patents: Firm-Level Evidence from China," with Jie Zhang and Yuan Zi, ***European Economic Review***, Vol. 133, April 2021, 103633.

[4] "Import and Innovation: Evidence from Chinese Firms," with Jie Zhang and Wenping Zheng, ***European Economic Review***, 2017 (94), 205–220.

[5] "Micro-to-Macro Gains from Regional Trade Agreements: Evidence from RCEP," with Zhihao Wang and Fuxin Zhai, *The World Economy*, forthcoming.

[6] "The Impact of Digitalization on Productivity: Evidence from Chinese Manufacturing Firms," with Hongxi Chen, Jingtao Yi, and Jinqiu He, *Management and Organization Review*, forthcoming.

[7] "Bank Competition, Defaulting Risks, and Innovation," with Mengbo Zhang, Aomeng Zhang, and Xinyang Li, *Canadian Journal of Economics*, forthcoming.

[8] "Trade Policy Uncertainty and Market Diversification by Risk-Averse Firms," with Zhihao Wang, Ting Zhu, Kejian Gu, and Ying Tang, *China Economic Review*, 2025, Vol. 91, 102400.

[9] "How Does Banking Expansion Influence City Development and Synergy? A New Perspective from Government Debts," with Yuyuan Wen and Hao Yu, *Cities*, 2025, 162, 105923.

[10] "Made and Created in China: The Role of Processing Trade," with Aksel Erbahar and Yuan Zi, *Scandinavian Journal of Economics*, 2025, 127, 390–426.

[11] "Detecting Learning from Importing: The Effect of Importing Behavior on R&D Expenditures of Chinese Firms," with Jingtao Yi and Jinchao Huang, *The World Economy*, 2024, 47(7), 3244–3278.

[12] "Is Cracking Down on Corruption Really Good for the Economy? Firm-Level Evidence from China," with Xin Jin and Xu Xu, *Journal of Law, Economics, and Organization*, 2021, 37(2), 314–357.

[13] "Types of Patents and Driving Forces behind the Patent Growth in China," with Jie Zhang, *Economic Modelling*, 2019 (80), 294–302.

[14] "The Bank–Firm Relationship: Helping or Grabbing?" with Yong Li and Jie Zhang, *International Review of Economics & Finance*, 2016 (42), 385–403.

Working Papers

[1] "Externalities and Vertical Bargaining in Two-Sided Retail Platforms," with Rong Luo and Mark Roberts.

[2] "International Cooperation of Trade and Industrial Policy," with Zhihao Wang and Wei Xiang.

[3] "The Uneven Reach of Skill-Biased Technology," with Xiangyu Feng and Zexuan Liu.

[4] "Do Local Industrial Policies Drive Battery Energy Storage Innovation?" with Xiangyang Zhao, Fuxin Zhai, Lun Li, and Shengyu Tao.

[5] "Detecting the Causal Impact of Strategic Decisions on Firm Performance," with Han Jiang, Moyu Liao, and Hao Sun.

[6] "Financing Innovation with Innovation," with Minjie Deng, Min Fang, and Yunzhou Shang, forthcoming.

[7] "Path-Length-Independent No-Regret Learning for Distributed Online Generalized Nash Games over Unbalanced Networks," with Zhenhua Deng and Yulan Qian.

[8] "User Rating and App Updates," with Jingtao Yi, Sihan Meng, Sali Li, and Noman Shaheer.

[9] "Learning and Scaling," with Zexuan Liu and Liang Chen, revision requested at ***Strategic Management Journal***.

[10] "Intellectual Property Financing, Innovation, and Aggregate TFP," with Guanliang Hu, resubmitted to ***Journal of Financial Economics***.

[11] "Data-Driven Vertical Integration," with Zexuan Liu, Fuxin Zhai, and Yao-yu Chih, revision requested at ***Management Science***.

[12] "Identification of Multiproduct Production Functions: Non-Parametric Limits and a CES Solution," with Moyu Liao and Jie Zhang, resubmitted at *Econometric Reviews*.

[13] "How Heterogeneous Are Productivity Gains from Exporting?" with Moyu Liao, Hao Yu, and Jinqiu He, revision requested at *Journal of Productivity Analysis*.

Selected Publications in Chinese Journals (English translation; original title in Chinese)

[1] "Measuring the Quality of Chinese Patents: Evolution Mechanisms and Economic Growth Effects," 中国专利质量的测度，变化机制与经济增长效应, with Yiran Zhang, Jie Zhang, and Zexuan Liu, *Economic Research Journal* （《经济研究》）, 2026(3).

[2] "Industrial Policy under the Integration of Innovation and Industrial Chains," 创新链与产业链融合下的产业政策, with Jie Zhang, Hao Sun, and Yuanming Ren, *Economic Research Journal* （《经济研究》）, 2024(9), 154–172. [Cover article]

[3] "Imported Foreign Technology and Indigenous Innovation of Chinese Firms," 对外技术引进与中国本土企业自主创新, with Jie Zhang, Shufeng Wu, and Wenhao Sun, *Economic Research Journal* （《经济研究》）, 2020(7), 92–105.

[4] "Evaluating China's Innovation Subsidy Policy: Theory and Evidence," 中国企业创新补贴政策绩效评估：理论与证据, with Jie Zhang, Fu Xin, and Lianxing Yang, *Economic Research Journal* （《经济研究》）, 2015(10), 4–17. [Lead article]

[5] "Measuring and Understanding Domestic Value Added in China's Exports," 中国出口国内附加值的测算与变化机制, with Jie Zhang and Yuanchun Liu, *Economic Research Journal* （《经济研究》）, 2013(10), 124–137. [Cover article]

[6] "Administrative Boundary Effects on Localized Knowledge Spillovers in China," 中国知识溢出本地化的行政边界效应研究, with Hao Yu and Jie Zhang, *China Industrial Economics* （《中国工业经济》）, 2025(1), 81–99.

[7] "Biased Technical Progress Effects of Tax Incentives: Evidence from the Accelerated Depreciation Policy," 减税激励的有偏技术进步效应研究——来自固定资产加速折旧政策的证据, with Yunzhou Shang and Jun Cui, *Statistical Research* （《统计研究》）, 2026(1).

[8] "High-Tech Enterprise Certification Policy and Firm Innovation," 高新技术企业认定政策与企业创新, with Wenkai Wang, *China Economics* （《中国经济学》）, 2024(2), 92–118.

[9] "Estimating and Decomposing the Effects of Patents on Firms' Revenue Productivity," 专利对企业报酬生产率的作用效应估计与分解, *China Journal of Econometrics* （《计量经济学报》）, 2023, 3(1), 166–194.

[10] "Manufacturing Agglomeration, Pollution, and Green Development: A Spatial Spillover Study," 制造业集聚、污染与绿色发展实践路径：基于空间溢出模型的研究, with Xiaoshan Ding, Chong Han, and Hao Yu, *Statistical Research* （《统计研究》）, 2022(9), 46–61.

[11] "Trade Policy Uncertainty and Firms' Export Product Scope," 贸易政策不确定性与企业出口产品范围, with Ting Zhu and Kexin Meng, *Journal of International Trade* （《国际贸易问题》）, 2022(6), 90–105.

[12] "A New Test of the Export–Productivity Relationship: Evidence from China," 出口与生产率关系的新检验：中国经验, with Jie Zhang and Fan Zhang, *Journal of World Economy* （《世界经济》）, 2016(6), 54–76.

[13] "Does Importing Induce Exporting? A Micro Interpretation of China's Export Miracle," 进口是否引致了出口：中国出口奇迹的微观解读, with Jie Zhang, Wenping Zheng, and Yujian Wang, *Journal of World Economy* （《世界经济》）, 2014(6). [Lead article]

[14] "Financing Constraints, Financing Channels, and Firms' R&D Investment," 融资约束、融资渠道与企业研发投入, with Jie Zhang, Wenping Zheng, and Zhe Lu, *Journal of World Economy* （《世界经济》）, 2012(10), 66–90.

Book

Structural Estimation of Production Functions and Applications （《生产函数的结构估计与应用》）, Economic Management Press, June 2026, ISBN 978-7-5243-1075-4.

Grants and Research Projects

NSFC General Program: "Production Function Identification and Measurement of Skill-Biased Technological Change in the Digital Economy," Principal Investigator. 2027–2030

NSFC Special Program: "Data Systems of Basic Research Supporting Economic Development," Subproject Leader. 2026–

NSFC Young Scientists Fund: "Measuring the Quality of Chinese Patents and Its Evolution Mechanisms Based on Patent Text Big Data," Principal Investigator. 2022–2024

Tencent Research Institute Project: "Patent Value, Innovation, and Technology Leadership of Platforms," Principal Investigator. 2023–2026

RUC Scientific Research Fund: "Quantitative Evaluation and Policy Practice of Patent Financing," Principal Investigator. 2023–2025

RUC Philosophy and Social Science Data Project: "China Patent Quality and Value Database," Principal Investigator. 2023

National Social Science Fund Major Project: "Accelerating the Construction of a Modernized Economic System," Subproject Leader. 2022

NSFC Key Project: "China's Innovation Incentive Mechanism with Synergistic Government and Market Functions," Subproject Leader. 2021

Professional Activities

Teaching at Renmin University of China

Empirical Methods in Economics and Business Studies (Master's/Ph.D.)	Fall 2020; Spring 2022–2026
Machine Learning and Causal Inference (undergraduate)	Spring 2026
Innovation Economics (undergraduate)	Fall 2023, 2024; Spring 2025, 2026
Topics on Industrial Policy (graduate)	Spring 2022–2024
Managerial Economics, iMBA (taught in English)	Fall 2022–2024; Spring 2025
Managerial Economics, MBA (taught in Chinese)	Fall 2023, 2024; Spring 2025
Global Network Week (GNAM) program	Spring 2024–2026
Market Analysis and Research (undergraduate)	Spring 2021

Teaching at Pennsylvania State University

Instructor, Introductory Macroeconomics (Econ 104)	Summer 2019
Instructor, Introductory Econometrics (Econ 306)	Summer 2018
Teaching Assistant, Intermediate Microeconomics	2017–2019
Summer Research Intern, Banco de México (Central Bank of Mexico)	2017/07–2017/09

Other Teaching Materials

Teaching cases published with the RUC Business School Case Center, 2023 and 2025.
Textbooks in preparation: *AI-Assisted Causal Inference and Machine Learning*; *Innovation Economics: Theory and Policy*, with Jie Zhang.

Referee Services

Abacus, Applied Economics Letters, Asian Journal of Technology Innovation, China Economic Review, China & World Economy, Economic Modelling, Economics Letters, Emerging Markets Finance and Trade, International Economics and Economic Policy, International Journal of Industrial Organization, International Review of Economics and Finance, Journal of Comparative Economics, Journal of Development Economics, Journal of International Management, Journal of Management Science and Engineering, Review of World Economy, The World Economy.
Also referee for Chinese journals including *Economic Research Journal* （《经济研究》）, *Social Sciences in China* （《中国社会科学》）, and *Journal of Financial Research* （《金融研究》）; conference referee for the Academy of International Business (AIB), 2024.

Conferences, Workshops, and Invited Seminars

* Co-author presented. Internal reading groups omitted.

2026:

- Invited lecture series *AI-Empowered One-Stop Research*: School of Economics, Renmin University (6/4); School of International Trade and Economics, Central University of Finance and Economics (6/9, 7/7–8); School of Economics, Nanjing University (6/15); WTO Research Institute, UIBE (6/17); School of Finance (6/24) and International School of Economics and Management (6/24), Capital University of Economics and Business; Business School, Beijing Normal University (7/9); China Center for Special Economic Zone Research, Shenzhen University (7/14).
- YES Summer Academic Lecture, Northwest University, 7/17.
- 10th CCER Summer Institute, National School of Development, Peking University, Chongli, 7/3–5.
- China Annual Conference on Digital Economy Development and Governance, Tsinghua University, 7/4.
- China Financial Research Conference (CFRC), PBC School of Finance, Tsinghua University, 7/2.
- China Economic Research Lecture Series and Symposium (1st), invited speaker, School of Economics, Renmin University, 1/20.

2025:

- 4th Conference on Open Economy and Macro Finance, School of Finance, Shanghai University of Finance and Economics, 12/27.

- 2nd Xiangzhang Academy Academic Lecture, keynote speaker, Business School, Hunan University of Science and Technology, 12/19.
- Invited seminar, Business School, Hunan Normal University, 12/18.
- 5th Xiangzhang Industrial Economics Forum, keynote speaker, Anhui University of Technology, 11/15.
- Cross-Strait Microeconomics Workshop: Theory and Applications, Xiamen University, 9/27–28.
- Invited seminar, Faculty of Social Sciences, University of Macau, 9/3.
- Eaton's Forum, International Economics and Finance Society, China, UIBE, 8/14.
- Young Scholars Academic Salon and 2nd Youth Development Economics Forum, Shandong University, 7/7–8.
- 3rd Global Alumni Academic Annual Conference, School of Finance, Renmin University, 7/4.
- ISEM Conference: Frontiers of Empirical Research in China, Capital University of Economics and Business, 5/19.

2024:

- Forum of the Alliance for Business Administration Disciplines Constructing China's Independent Knowledge System, "New Quality Productive Forces and Digital Economy" session; release of the *Platform Enterprise Digital Technology Invention Patent Research Report*, RUC Business School, 4/21.

2023:

- Penn State Alumni Empirical IO Workshop, School of Economics, Renmin University, 7/6.
- China Financial Research Conference, PBC School of Finance, Tsinghua University, 7/2.
- Invited seminar, Business School, Nanjing University, 4/7.

2022:

- Invited seminar, National Academy of Development and Strategy, Renmin University, 12/14.
- Invited seminar, Global Value Chain Research Institute, UIBE, 10/27.
- Invited seminar, China Center for Economic Studies, Fudan University, 10/11.
- 9th International Symposium on New Structural Economics, discussant, Institute of New Structural Economics, Peking University, 8/2.
- 6th CCER Summer Institute, National School of Development, Peking University, 7/2.
- 2022 Asian Meeting of the Econometric Society, CUHK-Shenzhen, 6/22.

2021:

- Invited seminar, School of Economics, Peking University, 11/19.
- 7th International Young Finance Scholars' Conference, Peking University HSBC Business School, 7/12.

2020:

- Invited seminar, Business School, Renmin University, 9/28.

2019:

- Invited seminars, School of International Trade and Economics, UIBE (12/12); Business School, Xiangtan University (12/10); School of Economics and Trade, Hunan University (12/4); CUHK-Shenzhen; Institute of New Structural Economics, Peking University; School of Economics, Renmin University.
- The 14th Annual Economics Graduate Student Conference, Washington University in St. Louis.
- *Midwest International Economic Development Conference.

2018:

- *The first CICE Annual Conference, Tsinghua University.
- AEA Annual Meeting, poster session, Philadelphia.
- *Workshop in Applied and Theoretical Economics (WATE).
- *Special Issue Conference on "The Challenges of Managing and Modelling Innovation and Growth in China," Renmin University.

2017:

- China Meeting of the Econometric Society, Wuhan.
- *Annual Meeting of the International Consortium for China Studies, Mannheim.

2014: Annual Geneva–China Workshop, Geneva.

2013: Doctoral Forum, Fudan University.

Honors and Awards

National Youth Talent Program, 2025
Highly Cited Scholar (Top 1%), Economics, CNKI, 2025 and 2024

The 7th Peikang Chang Award for Outstanding Papers in Development Economics, Peikang Chang Research Foundation of Development Economics, 2018

The 7th Higher School Outstanding Scientific Research Achievement Award (Humanities and Social Sciences), Third Prize, Ministry of Education

Top 10 Chinese Papers on International Trade in 2016, Institute of World Economics and Politics, Chinese Academy of Social Sciences, 2017

The Third Prize of the 18th An Zijie International Trade Research Awards, UIBE, 2014

National Scholarship (Top 1%), Renmin University of China, 2012, 2013, and 2014

Patents, Software, and Skills

Patent: "Method and System for Patent Quality Measurement Based on Patent Text Similarity" （基于专利文本相似度的专利质量测算方法及系统） (application under substantive examination), with Zexuan Liu and Xiangyang Zhao.

Software: Author of `pte`, a Stata module for productivity treatment effects estimation (with X. Cai, M. Liao, and W. Xu), 2026. Proficient in MATLAB, Stata, Julia, and Python.

Language: Chinese (native), English (fluent)